NOISE AND VIBRATION ANALYSIS

SIGNAL ANALYSIS AND EXPERIMENTAL PROCEDURES

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Single-input Frequency Response Measurements

It is common in many noise and vibration applications to compute frequency response functions from measurements. In most cases, the structure is excited by known (measured) forces applied either by an impulse hammer or by a shaker. In some cases, frequency response functions are measured between response signals which are due to natural excitation by for example wind or traffic loads on buildings or bridges. The latter is common, e.g., when measuring transmissibilities for operating deflection shape measurements, see Section 16.6.

Common reasons that one may wish to measure frequency response functions between a force and a response signal are, for example, to determine

- natural frequencies and relative damping (e.g., in experimental modal analysis, see Section 16.7)
- point stiffness to be used in analytical models
- noise transfer paths (NTF) for *noise path analysis* (NPA), which is a method of studying sound paths from, for example, the engine mounts to the driver's ear in a vehicle, see Chapter 15. NPA has so far mainly been used within the automotive industry, but is increasingly being applied for other products.

In this chapter, we shall present two different ways of measuring frequency response, namely through excitation with an impulse hammer and with a shaker, respectively. We limit the discussion in this chapter to systems with one input and one output, so-called single-input/single-output (SISO) systems. In Chapter 14 we will extend the concept to general systems with many inputs and many outputs. Much of the theory in this chapter was developed in the early days of noise and vibration analysis. A good source for a more detailed discussion is Bendat and Piersol (2010).

Estimation of frequency response turns out to be rather complicated in terms of all the errors involved, because, as we will see, there are errors due to the spectral estimation as well as errors due to unwanted noise in the measured force and response signals. We will therefore make extensive use of simulations in this chapter, rather than real measurements. This allows us to focus on one of the sources of error at the time, whereas in real measurements all errors occur at once, and are unknown. In order to make good measurements of FRFs, it is essential to understand all the errors involved, and to be able to determine when or if they are occurring in a particular measurement.

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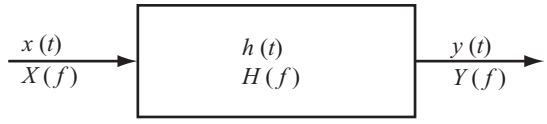


Figure 13.1 Linear time-invariant system with input $x(t)$, output $y(t)$, impulse response $h(t)$ and frequency response $H(f)$

13.1 Linear Systems

We recall from Section 2.6, that a system with an input signal $x(t)$ and an output $y(t)$ is considered to be *time invariant* if the coefficients of the system of differential equations which describe the system do not change with time. In practice, this means that the system parameters, for example mass, damping, and stiffness, do not change during the time we study (measure) the system. Further, a system is *linear* if it fulfills two criteria: (i) the system is *additive*; and (ii) the system is *homogeneous*.

Real-life systems are rarely completely linear. During all measurements of frequency response we therefore need to investigate whether or not the conditions of linearity are fulfilled. If they are not, then many of our common assumptions fail, and the methods presented in this chapter will not be appropriate. How to check the linearity of mechanical systems will be discussed in Section 16.7.2.

13.2 Determining Frequency Response Experimentally

We shall now see how the frequency response can be estimated experimentally, under the assumption that the system, $H(f)$, is linear and time-invariant. We thus assume that we have a system as in Figure 13.1, with input $x(t)$ and output $y(t)$. Recall from Section 2.6 that we may write the output as the convolution of the input with the system's impulse response, $h(t)$, i.e.,

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(u)h(t-u)\mathrm{d}u. \tag{13.1}$$

Recall also that we can carry out the computation of Equation (13.1) in the frequency domain, which is undeniably more attractive, since in that case we obtain the equation for the spectrum $Y(f)$ as

$$Y(f) = X(f) \cdot H(f). \tag{13.2}$$

The input is thus amplified and phase-shifted at each frequency independently of other frequencies.

In real-life measurements, particularly in the field of noise and vibrations, we often have no hope of being able to measure $x(t)$ and $y(t)$ without at least one of these signals being contaminated by extraneous noise from the sensor and input electronics in the measurement system. However, we can assume that the contaminating noise is uncorrelated with the input and output signals. A general model for an actual system where extraneous noise is added to the signals we measure is illustrated in Figure 13.2. A discussion about the contaminating noise on the input and output signals is found in Section 13.7.

13.2.1 Method 1 – the H_1 Estimator

In order to find a method which estimates the frequency response of the linear system, we shall first make a simplification of the system in Figure 13.2, which consists of setting the noise at the input to $m(t) = 0$. We thus assume that we can measure the input $x(t)$ without any extraneous noise, but that the output contains noise. Furthermore, we assume for the moment that the input signal, $x(t)$ is random noise. In Section 13.7.1 we will extend the discussion to input signals with other properties.

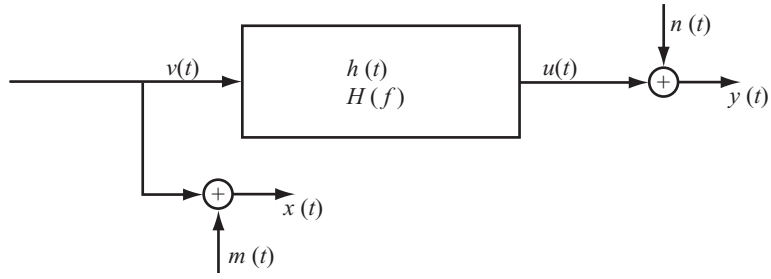


Figure 13.2 Linear system with extraneous noise contaminating both the input and output signals

For the system in Figure 13.3 we can formulate the spectrum of the output as

$$Y(f) = X(f)H(f) + N(f). \tag{13.3}$$

Next, we multiply both the left-hand and right-hand sides in Equation (13.3) by $X^*(f)$, i.e., the complex conjugate of $X(f)$, and obtain

$$X^*(f)Y(f) = X^*(f)X(f)H(f) + X^*(f)N(f). \tag{13.4}$$

If we take the expected values (i.e., experimentally we make an average) of the left-hand side and each term on the right-hand side separately, and scale the results in a suitable way to obtain PSDs as descried in Section 10.3, we obtain, from Equations (10.11) and (10.13), the cross- and auto-spectral densities

$$G_{yx}(f) = G_{xx}(f)H(f) + G_{nx}(f). \tag{13.5}$$

The last term in Equation (13.5), the cross-spectral density between the input and the noise, approaches zero when we average, since these signals are uncorrelated. We thus obtain the so-called H_1 *estimator* of $H(f)$, as

$$\hat{H}_1(f) = \frac{\hat{G}_{yx}(f)}{\hat{G}_{xx}(f)} \tag{13.6}$$

which is the equation used in FFT analysis systems for estimating $H(f)$ from measurements of $x(t)$ and $y(t)$ if it is assumed that the contaminating noise in the input signal is negligible. As before, we use the symbol^(hat) to denote that we are dealing with estimated functions. Note specially in Equation (13.6), that, because $G_{xx}(f)$ is real, the phase angle in $H_1(f)$ comes exclusively from the phase angle of the cross spectrum, $G_{yx}(f)$.

The H_1 estimator in Equation (13.6) is a *least squares* estimate of the system $H(f)$, see for example Bendat and Piersol (2010). If there is noise added to the input, and thus the model we use for the H_1 estimator is wrong, then the estimated H_1 will be *biased*. This will be discussed in Section 13.4. Furthermore, as we will see in Section 13.5.1 the H_1 estimator is always biased due to the limited frequency resolution, although this error can be made arbitrarily small by increasing the blocksize.

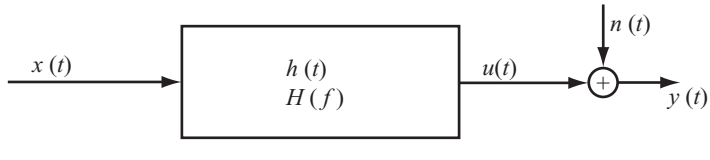


Figure 13.3 Linear system with noise only at the output

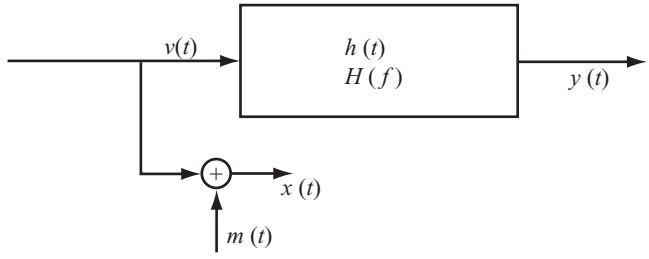


Figure 13.4 Linear system with noise only at the input

Example 13.2.1 *Write a MATLAB/Octave script which calculates the H_1 estimator of a single-input/single-output system, using spectral densities in variables* `Gxx` *and* `Gyx`.

We assume the variables are column vectors as described in Section 10.7.1. The H_1 estimator is then simply the element-wise division of the cross-spectral density and the autospectral density of x.

```
H1=Gyx./Gxx;
```

End of example.

13.2.2 Method 2 – the H_2 Estimator

In some cases it may be the case that the preceding assumption, that the noise at the input is negligible, is not reasonable. For example, when we measure the accelerance of mechanical systems with shaker excitation, then near the natural frequencies of the structure, the acceleration signal (output) is large, but the force signal (input) is often small because the structure is weak, see also Section 13.9. In that case it is more reasonable to assume that the dominating extraneous measurement noise is present in the input signal, $x(t)$, as illustrated in Figure 13.4.

To find an estimator for this case, from Figure 13.4 we obtain the following relationship

$$Y(f) = [X(f) - M(f)]\, H(f). \tag{13.7}$$

The trick this time is to multiply Equation (13.7) by the complex conjugate of the output spectrum, $Y^*(f)$, which gives

$$Y^*(f)Y(f) = H(f)\left[G_{xy}(f) - G_{my}(f)\right]. \tag{13.8}$$

We now take the expected value (average) like we did for the H_1 estimator above, and scale properly, and obtain

$$G_{yy}(f) = H(f)\left[G_{xy}(f) - G_{my}(f)\right]. \tag{13.9}$$

Similar to the H_1 case above, the cross-spectrum, $G_{my}(f)$ will approach zero when we average, if we assume that the noise, $m(t)$, is uncorrelated with $v(t)$ and therefore with $y(t)$. We thus obtain the so-called H_2 *estimator* as

$$\hat{H}_2(f) = \frac{\hat{G}_{yy}(f)}{\hat{G}_{xy}(f)} \tag{13.10}$$

Note in Equation (13.10) that $G_{xy}(f) = G^*_{yx}(f)$ (see e.g., Equation 8.13), that is, the phase of $G_{xy}(f)$ is equal to the phase of $G_{yx}(f)$ with opposite sign. Since $G_{xy}(f)$ in Equation (13.10) is in the denominator while $G_{yx}(f)$ is in the numerator in Equation (13.6), we see that the phase of H_1 is equal to the phase of

H_2. It should also be noted that if the H_2 estimator is missing in the FFT analysis software, it can easily be computed by switching place between x and y and then inverting the resulting H_1 estimate, i.e.,

$$^2\hat{H}_{yx} = \frac{1}{^1\hat{H}_{xy}}. \tag{13.11}$$

where we introduce the left superscript '1' in $^1\hat{H}_{yx}$, etc., to indicate that we are using specifically the H_1 estimator. This nomenclature will be used whenever it is not obvious from the context which particular estimator we are discussing.

Like the H_1 estimator in the case of noise on the output, the H_2 estimator is also a least squares estimate of $H(f)$. In case the model assumption is wrong, the estimate $\hat{H}_2$ is biased, which will be discussed in Section 13.4. It is also, like the H_1 estimator, always biased due to the limited frequency resolution, although this error can be made arbitrarily small by increasing the blocksize, see Section 13.5.1.

13.2.3 Method 3 – the H_c Estimator

Now, what if we are taking the noise on both input and output into consideration? There has been several attempts at solving this system, including the H_v (Rocklin *et al*. 1985), and H_s (Wicks and Vold 1986) estimators. In order to get an unbiased estimate in the case of noise on both input and output, it turns out that this is only possible with some *a priori* information about the input and output noise (or their ratio), or, by knowing one signal without any contaminating noise at all. In Section 14.1.6 we will look at the common H_v estimator which is an estimator trying to address this problem. It should also be emphasized that the discussion here applies to random input signals; in Section 13.12 we will show that we *can* remove the bias due to both input and output noise if we use a periodic excitation signal (or at least make it arbitrarily small).

Here, we will look at another, rather ingenious, estimator, namely the H_c estimator, which assumes that there is a signal which can be measured without contaminating noise. In the case of shaker excitation of structures, the electrical output signal of the signal generator is typically chosen. This estimator was originally proposed by Goyder (1984), and further developed by Mitchell and Cobb (1987).

In Figure 13.5, the concept of the H_c estimator setup is illustrated. We have a signal $v(t)$ which we assume we can measure without any contaminating noise. In addition we have the force signal, $x(t)$, and a response signal, $y(t)$, both of which are measured with contaminating noise, $m(t)$, and $n(t)$, respectively.

The H_c estimator is now based on the fact that, using the H_1 estimator, we know from Section 13.2.1 that we can estimate the two systems, $^1\hat{H}_{yv}$ and $^1\hat{H}_{xv}$, indicated in Figure 13.5 without bias due to the

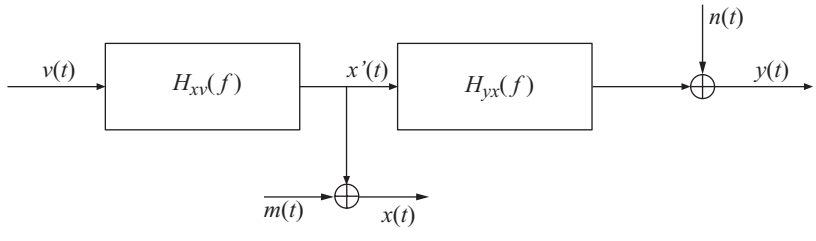


Figure 13.5 Illustration of the model assumption for the H_c estimator. The estimator is based on the assumption that there is an input signal, $v(t)$, which can be measured without any contaminating noise, and which is located before the measured input to the system. In mechanical applications, this is typically the voltage output of the signal generator

extraneous noise, if we just make appropriately many averages. But we now have the simple relation that the system we are seeking, H_{yx} is given by

$$H_{yx} = \frac{Y(f)}{X(f)} = \frac{Y(f)/V(f)}{X(f)/V(f)} = \frac{H_{yv}}{H_{xv}} \tag{13.12}$$

which means that we can define the H_c estimator by

$${}^{c}\hat{H}_{yx}(f) = \frac{{}^{1}\hat{H}_{yv}}{{}^{1}\hat{H}_{xv}}. \tag{13.13}$$

In case we make enough averages so that the H_1 estimators in Equation (13.13) are unbiased, then the H_c estimator is also unbiased. This is at the expense of an extra input channel, however, which is probably a reason why this estimator is not very common in commercial software. A further disadvantage with this estimator is that it relies on the frequency responses involved being linear, whereas for shakers used for structural excitation, we know they are not linear, which can cause uncertainty in the H_c estimator.

13.3 Important Relationships for Linear Systems

We shall now study some important relationships which apply to linear systems. If we measure the input $x(t)$ to a linear system with frequency response $H(f)$, the complex conjugate of Equation (13.2) is

$$Y^*(f) = X^*(f)H^*(f). \tag{13.14}$$

Multiplying Equation (13.2) by Equation (13.14) term by term, taking the expected value of each term, and scaling properly, we obtain the important relation

$$G_{yy}(f) = |H(f)|^2 G_{xx}(f) \tag{13.15}$$

This equation is a useful relation to express the PSD of the output of a linear system. Note that Equation (13.15) is valid for ideal linear systems without contaminating noise on the measured signals, as in Figure 13.1.

We now move to the system with noise only at the output, in Figure 13.3. If we start with the signals summed at the output we have that

$$Y(f) = U(f) + N(f). \tag{13.16}$$

We form the complex conjugate of this equation and multiply it by itself, obtaining

$$Y^*Y = U^*U + N^*N + U^*N + UN^* \tag{13.17}$$

where we have left out the frequency variable for the sake of simplicity. Taking the expected value of each term, and scaling properly, we obtain

$$G_{yy} = G_{uu} + G_{nn} + G_{nu} + G_{un}. \tag{13.18}$$

Equation (13.18) expresses a very important relationship, namely that *if we sum several signals, the power spectral density of the summed signal is in general not equal to the sum of each signal's PSD*. Instead, we must take into account all of the cross spectral densities between the signals included in the summation.

If the signals u and n are *uncorrelated* (independent), however, which they are in our case for the system with noise at the output, we obtain the special case

$$G_{yy} = G_{uu} + G_{nn}. \tag{13.19}$$

13.4 The Coherence Function

In the case of existing input noise, $m(t)$, not included in the assumption for the H_1 estimator, the magnitude of the estimate $\hat{H}_1$ will be less than or equal to the true value of $H(f)$. This can easily be observed by the fact that when $m(t)$ is not zero, then $\hat{G}_{xx}$ will be measured with this noise and thus

$$\mathrm{E}\left[\hat{G}_{xx}\right] = G_{vv} + G_{mm} \tag{13.20}$$

according to Equation (13.19). Because the $\hat{G}_{xx}$ term is found in the denominator of the H_1 estimator in Equation (13.6), $\left|\hat{H}_1\right|$ will become *smaller than or equal to* the true $|H(f)|$ due to this bias error. In a similar fashion, because the H_2 estimator has a factor G_{yy} in the numerator which will be affected similarly with G_{nn}, $\left|\hat{H}_2\right|$ will always be *greater than or equal* to the true value $H(f)$, and equal only when $n = 0$. In both cases it is also necessary that we have made many averages in Equation (13.6) and (13.10), respectively, so that the cross-spectrum terms, including the extraneous noise in Equations (13.5) and (13.9), respectively, become zero. Thus, we may conclude that the true value of $|H(f)|$ fulfills

$$\left|\hat{H}_1(f)\right| \leq |H(f)| \leq \left|\hat{H}_2(f)\right|. \tag{13.21}$$

When we estimate the frequency response with the H_1 or H_2 estimators as above, we can simultaneously compute the *coherence function*, $\gamma^2(f)$ which is defined as the ratio between the $\hat{H}_1$ estimate and the $\hat{H}_2$ estimate, i.e.,

$$\hat{\gamma}^2_{yx}(f) = \frac{\hat{H}_1(f)}{\hat{H}_2(f)} = \frac{\left|\hat{G}_{yx}(f)\right|^2}{\hat{G}_{xx}(f)\hat{G}_{yy}(f)}. \tag{13.22}$$

Note that the coherence function in Equation (13.22) is defined as the *squared* function, γ^2_{yx}. From Equation (13.21) and the definition of the coherence function, it follows directly that

$$0 \leq \gamma^2_{yx}(f) \leq 1. \tag{13.23}$$

If $\gamma^2_{yx}(f) = 1$ then $\hat{H}_1 = \hat{H}_2$ which implies that we have no extraneous noise, and moreover that the measured output, $y(t)$, derives solely from the measured input, $x(t)$.

The coherence function is a quality measure of our estimated frequency response, regardless of which estimator we use. The coherence function drops below unity if there is contaminating noise on either the measured input signal, $x(t)$, or in the measured output signal, $y(t)$, or in both signals. In all three cases, there is a bias error in the determination of the frequency response in at least one of the estimators, $\hat{H}_1$, or $\hat{H}_2$, which follows from the fact that $\hat{\gamma}^2_{yx} = \hat{H}_1/\hat{H}_2$ and the properties in Equation (13.21). It the coherence drops because of noise $m(t)$ or $n(t)$ (or both, of course) is impossible to tell solely from observing $x(t)$ and $y(t)$. We will return to a more thorough discussion on the coherence function in Section 13.7, but first we need to establish some more relations.

13.5 Errors in Determining the Frequency Response

We will now look at some aspects of the errors involved in estimating frequency response functions using the H_1 and H_2 estimators for a single-input/single-output system. This is considerably more complicated than the discussion on spectrum estimation in Chapter 9, because the correlation between the input and output signals is coming into the error relations. Also, the estimators we use to estimate the auto- and cross-spectral densities change the relations for the bias and random errors involved.

It is important to understand that there are two completely different errors involved in the FRF estimates: (i) spectral analysis errors, and (ii) model errors. We have already discussed the model errors, which arise if, for example, we use the H_1 estimator, but there is input noise, $m(t)$. The errors we will look at in this section assume that the model is correct.

The spectral errors in the FRF estimates are further divided into two parts; the errors caused by the estimator itself, without any extraneous noise, $n(t)$ or $m(t)$ as illustrated in Figure 13.2, and then the errors caused by the extraneous noise, i.e., $n(t)$ if we look at the H_1 estimator, and $m(t)$ if we look at the H_2 estimator.

The most common estimator for the auto- and cross-spectral densities, used in all commercial systems to date, is Welch's method, as described in Section 10.3.2. A comprehensive analysis of the errors involved when using this method when the input signal is random noise, is found in Antoni and Schoukens (2007, 2009) and Schoukens *et al.* (2006). A thorough discussion on errors if the smoothed periodogram estimator is used is found in Bendat and Piersol (2010).

13.5.1 Bias Error in FRF Estimates

Similarly to what we discussed for spectral densities in, e.g., Section 10.3.4, the frequency response estimated with the H_1 estimator, even without the presence of contaminating noise, is biased due to the limited frequency resolution. In practical frequency response measurements, this bias error should be minimized by selecting a blocksize for the FFT by using the same procedure as presented in Section 10.6.3, i.e., by gradually increasing the blocksize until peaks do not increase in height when the blocksize is increased further.

The spectral analysis bias error has two parts; one caused by the fact that the measured output block is not actually caused exactly by the measured input block, and one caused by leakage. The first of these errors is illustrated in Figure 13.6 and is caused by the fact that when we truncate the signals, we measure

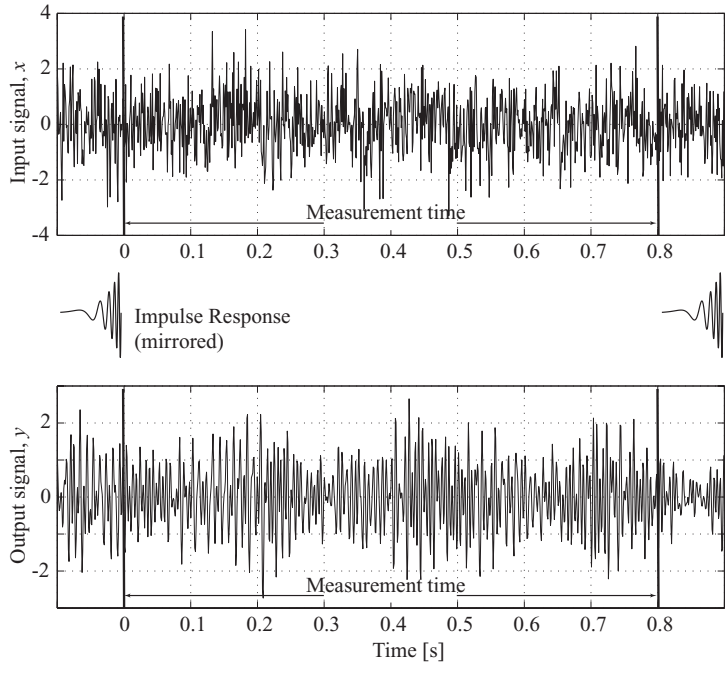


Figure 13.6 Illustration of an arbitrary time block of the input and output signals in the averaging process to compute a frequency response function. A 'time mirrored' impulse response is added to the picture to illustrate the convolution of the input signal with the impulse response, which involves mirroring the impulse response before multiplying the input signal and summing, see Chapter 2. The measured output block is therefore not caused by (exactly) the measured input block, which causes an error. This error is minimized by increasing the measurement time so that it is much larger than the length of the impulse response

an output signal of the linear system, which is not totally caused by the input signal. To understand this effect, we must consider that the output signal is the input signal convolved with the input signal, which from Section 2.6.4 we know means that the impulse response is 'mirrored' (time reversed), then multiplied by the input signal, and the product is summed up. At the beginning of the output signal in a particular time block, there is, therefore, a region which actually depends on the input signal *prior* to the time block. At the end of the time block, there is correspondingly a part of the input signal that causes the output signal outside (after) the time block. This error is often erroneously called leakage, although it is not really the same as the previously mentioned effect called leakage in Chapter 9 which was due to the end effects of the blocks in the DFT process.

Once the blocksize is made much larger than the impulse response length, the remaining bias error is caused by leakage. An important result in Antoni and Schoukens (2007) is that the normalized bias error, $\varepsilon_b\left[\hat{H}\right]$, of a frequency response estimate is proportional to

$$\varepsilon_b\left[\hat{H}\right] \propto \frac{1}{N^2}\frac{H''(f)}{H(f)} \tag{13.24}$$

where H'' is the second derivative of the FRF with respect to frequency. First of all, we can conclude that the bias error is inversely proportional to the blocksize squared, which means it vanishes quickly as we make the blocksize large. In Section 13.10 we will illustrate how this bias error can be made negligible in measurements of a typical lightly damped system.

We recognize the ratio of the second derivative to the value of the FRF from the bias error expression for spectral densities in Equation (10.18). Since we usually measure FRFs of lightly damped systems in noise and vibration analysis applications, it is of particular interest to know the nature of the bias error on such systems. It turns out that the bias error, similarly to the spectral density bias error, essentially depends on the ratio of the resonance bandwidth and the frequency increment, i.e., the ratio $B_r/\Delta f$. A plot of a simulation result of the maximum bias error (located at the resonance frequency) of a frequency response of an SDOF system, is presented in Figure 13.7.

Example 13.5.1 *Assume we have a truck frame with a first natural frequency of 1 Hz, and relative damping of 1%. Determine the necessary frequency increment so that the normalized bias error of an FRF, measured using random excitation, is below 1%. If you need to make 50 independent averages (without overlapping), how long will the measurement take?*

The resonance bandwidth according to Equation (5.39) is $B_r = 2\zeta_r f_r = 0.02$. *In order to have a bias error less than 1%, we need to have a frequency increment of maximum* $\Delta f_{max} = 0.1B_r$ *according to the plot in Figure 13.7. Thus, we need to have a frequency increment of less than 0.002 Hz. Since the time of each FFT time block is* $T = 1/\Delta f$ *this means the measurement will take*

$$T_m = \frac{50}{0.002} = 25000s \approx 7\,hours. \tag{13.25}$$

This example shows what a high price low-bias measurements of FRFs can come at!

End of example.

13.5.2 Random Error in FRF Estimates

As shown by Antoni and Schoukens (2007) and (Schoukens *et al.* (2006), the random error in estimates of frequency responses using Welch's method, has two parts; one caused by the extraneous noise, $n(t)$ in the H_1 case, and one caused by leakage, of which the latter has previously been unaccounted for. The total random error therefore depends on the time window as well as the number of averages and the overlap percentage, similarly (but not equivalently) with the random error of PSD estimates. In Antoni and Schoukens (2007) it was shown that the smallest random error possible, given a certain amount of data, is obtained by using a *half sine window* instead of the more common Hanning window, and with an

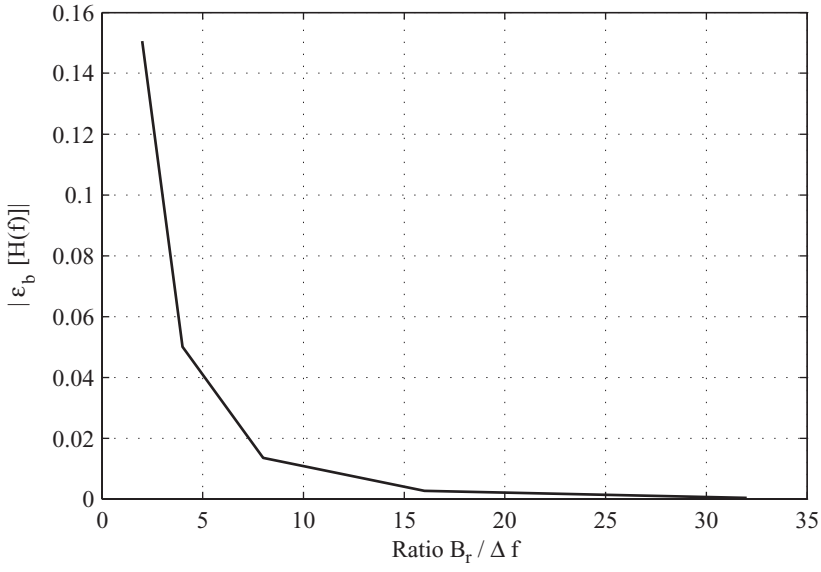


Figure 13.7 Maximum normalized bias error of FRF estimate on an SDOF system, versus the ratio of the resonance bandwidth and the frequency increment, $B_r/\Delta f$. The plot is a result of a simulation of the maximum bias error, which occurs at the resonance frequency, using various blocksizes. Note that the plot shows the magnitude of the bias error; the bias error is actually negative at the resonance frequency. Furthermore, the plot was obtained by estimating the FRF using a half sine window and 67% overlap, see Section 13.5.2. The plot shows that in order to have a normalized bias error of less than 1%, we have to have at least ten frequency lines within the resonance bandwidth. It also shows, however as does Equation (13.24), that the bias error can be made arbitrarily small by increasing the blocksize

overlap percentage of 67% instead of the more common overlap of 50% used for spectrum analysis. The differences in results using a Hanning and a half sine window are not very large, but it is always good practice to use the best possible method. This will be further discussed and illustrated in Section 13.9. It should be noted that the half sine window is unsuitable for spectrum estimation, so if both spectra and frequency responses are to be calculated and used, it may be better to use the common Hanning window.

The random errors as defined by Antoni and Schoukens (2007) are rather complicated to interpret. For our purpose it is sufficient to provide an approximate equation for the random error in the magnitude of frequency response functions, given by Bendat and Piersol (2010). They give the random error of the frequency response between input $x(t)$ and output $y(t)$ for the H_1 estimator, in the absence of extraneous noise on the input, as

$$\varepsilon_r\left[\left|\hat{H}(f)\right|\right] \approx \sqrt{\frac{1-\gamma_{xy}^2(f)}{2n_d\gamma_{xy}^2(f)}} \tag{13.26}$$

where n_d is the number of distinct averages, i.e., the number of averages without overlap processing. When using overlap processing, n_d should be replaced by M_e, the equivalent number of averages, as given by Figure 10.8. This equation, although neglecting the random error due to leakage, is attractive in its simplicity, as it reveals that the best way to achieve a small random error is to make sure the coherence is near unity. If the coherence is not near unity, due to excessive output noise, the random error in the estimate can still be reduced by a (large) number of averages. A near unity coherence function should always be attempted to achieve when measuring frequency responses, as will be demonstrated

in Section 13.9. In addition, it should be mentioned that in practice we usually also have other errors, such as extraneous noise on the input signal, which makes the precision of this equation good enough for practical use.

13.5.3 Bias and Random Error Trade-offs

From Sections 13.5.1 and 13.5.2 it can be concluded that the bias error and random error are contradictory, as they were shown to be for spectral density estimates in Chapter 10. To reduce the bias error we need to use large blocksize, which (if we have a fixed amount of recorded data) reduces the possible number of averages. To reduce the random error we need to take many averages, which (under the premise of a fixed amount of data) makes the available blocksize less. We thus have to make a compromise between bias and random errors, if we have a certain amount of data. In practical FRF measurements, however, it is good practice to experimentally verify a sufficiently small bias error, prior to recording the data, and then to verify how many averages are needed to produce a sufficiently small random error. In many cases it is necessary to analyze the necessary measurement time prior to the measurement, so that enough averages can be made to get a reasonably low random error and bias error.

Instead of using Welch's method to compute the auto- and cross-spectral densities needed for the FRF estimator, it is, of course, possible to use the smoothed periodogram method described in Section 10.3.6. Actually, this estimator suffers a little less from the end block leakage than Welch's method, particularly for small blocksizes. However, the difference is so small that from a practical point of view it is equivalent to the Welch estimator. It can also be shown that the two estimators are asymptotically equivalent when the blocksize increases, (Antoni and Schoukens 2007).

13.6 Coherent Output Power

Assuming we have a system with noise on the output only, as in Figure 13.3, we will now look at some properties which allow us to compute the spectral densities of the output from the linear system, G_{uu}, and of the extraneous noise, G_{nn}. From Equation (13.15) and the expression for $H(f)$ with the H_1 estimator from Equation (13.6) we first have

$$\hat{G}_{uu} = \left|{}^{1}\hat{H}\right|^2 \hat{G}_{xx} = \frac{\left|\hat{G}_{yx}\right|^2}{\hat{G}_{xx}}. \tag{13.27}$$

Using the first and second term in this equation we can thus estimate the spectral density of $u(t)$, i.e., the part of $y(t)$ which comes from $x(t)$ through the linear system, even though we cannot measure $u(t)$.

By noting the similarity between Equation (13.27) and the definition of the coherence function in Equation (13.22), we also find that Equation (13.27) can alternatively be calculated as

$$\hat{G}_{uu}(f) = \hat{\gamma}^2(f) \cdot \hat{G}_{yy}(f) \tag{13.28}$$

which is the usual formula used to compute $\hat{G}_{uu}$.

$\hat{G}_{uu}(f)$ is called the *coherent output power (spectrum)*, since it stands for the part of $G_{yy}(f)$ which linearly derives from the input, $x(t)$, i.e., which is *coherent with* $x(t)$. This function can be used to identify noise sources (Bendat and Piersol 1993). However, it is crucial to remember that this formula is based on the system without noise at the input. In order to identify noise sources using Equation (13.28) it is thus important to find a 'clean' input, which can be measured without extraneous noise, see Section 15.3.

Using Equation (13.28) we can also express Equation (13.19) as

$$\hat{G}_{nn}(f) = \left(1 - \hat{\gamma}^2_{yx}(f)\right) \hat{G}_{yy}(f) \tag{13.29}$$

because x (and therefore u) and n are uncorrelated.

We can therefore also estimate the PSD of the contaminating noise, even though we cannot measure this signal. Again, this formula is valid under the assumption that no noise exists at the input of the system.

13.7 The Coherence Function in Practice

We will now find a few more relations which help in interpreting the coherence function. First of all we should stress that the coherence function is a result of the fact that there is a difference between the H_1 and the H_2 estimator results, i.e.,

$$\mathrm{E}\left[\frac{\hat{G}_{yx}}{\hat{G}_{xx}}\right] \neq \mathrm{E}\left[\frac{\hat{G}_{yy}}{\hat{G}_{xy}}\right]. \tag{13.30}$$

Also, it is apparent that the coherence function equals exactly 1 if we only make one average, since then

$$\hat{\gamma}^2_{yx} = \frac{\left|\hat{G}_{yx}\right|^2}{\hat{G}_{xx}\hat{G}_{yy}} = \frac{(YX^*)(Y^*X)}{XX^*YY^*} = 1. \tag{13.31}$$

This should be interpreted such that the coherence is *undefined* when only one average is taken.

In practice, the coherence function requires more averages to obtain a small random error, than does the frequency response. There is no exact solution of the random error in coherence estimates, but Bendat and Piersol (2000) give an approximate equation where the normalized random error is given by

$$\varepsilon\left[\hat{\gamma}^2_{yx}\right] \approx \frac{\sqrt{2}(1-\gamma^2_{yx})}{\left|\gamma_{yx}\right|\sqrt{n_d}} \tag{13.32}$$

where n_d is the number of distinct averages, without overlap processing. It is worth pointing out that this error formula means that a very large number of averages is needed to keep the random error low when the coherence is low. For example, a random error of $\varepsilon \leq 0.02$, when the coherence is $\gamma^2_y x = 0.5$, requires 5000 distinct averages, whereas when the coherence is $\gamma^2_{yx} = 0.8$, only 500 averages is needed to produce the same random error. With overlap processing, n_d in Equation (13.32) should be replaced by M_e, the apparent number of averages from Figure 10.8. It is apparent that in general the error given by Equation (13.32) is larger than the random error of the magnitude in the FRF, given by Equation (13.26).

We will now look at the quantitative value of the coherence function, in the cases of noise on the output and input, respectively. If we assume a system with extraneous noise on the output only, combining Equation (13.19) and Equation (13.28) leads to the relation

$$\gamma^2_{yx} = \frac{G_{uu}}{G_{yy}} = \frac{G_{uu}}{G_{uu}+G_{nn}} = \frac{1}{1+\frac{G_{nn}}{G_{uu}}}. \tag{13.33}$$

From this equation it follows that the coherence function deviates from unity when the extraneous output noise is not zero.

If, instead, we assume a system with noise on the input only, then according to Figure 13.4 we have that

$$G_{yx} = G_{yv} = {}^2HG_{vv} \tag{13.34}$$

which follows if we formulate the equation for Y as a function of V, multiply by X^* and then average over many blocks. But we also have, from Equation (13.27),

$$G_{yy} = \left|{}^2H\right|^2 G_{vv}. \tag{13.35}$$

If we combine the two last equations with Equation (13.22) we arrive at the relation

$$\gamma_{yx}^2 = \frac{\left|G_{yx}\right|^2}{G_{xx}G_{yy}} = \frac{\left|G_{yv}\right|^2}{(G_{vv}+G_{mm})\,G_{yy}} = \frac{\left|^2H\right|^2 G_{vv}^2}{(G_{vv}+G_{mm})\left|^2H\right|^2 G_{vv}} = \frac{1}{1+\frac{G_{mm}}{G_{vv}}}. \tag{13.36}$$

Equation (13.36) shows how the coherence deviates from unity when there is noise (only) on the input. It should be noted that the coherence function is thus, by combining Equations (13.33) and (13.36), a result of the amount of total extraneous input and output noise in the measurement in the general case of noise on both input and output signals.

We can now summarize some possible reasons for the coherence function to deviate from unity:

- Either the noise $m(t)$ or $n(t)$, or both, are not negligible compared with the measured signals $x(t)$ and $y(t)$.
- Bias errors due to insufficient frequency resolution, as described in Section 13.5.1.
- The system $H(f)$ is (strongly) nonlinear or not time invariant.
- Bias error exist due to a time delay between the signals $x(t)$ and $y(t)$. This is explained in detail in Bendat and Piersol (2010).
- The model is incorrect, for example, $v(t)$ is not the only (correlated) input which contributes to $y(t)$. This is explained in detail in Bendat and Piersol (2010).

13.7.1 Non-random Excitation

If the input signal $x(t)$ to a linear system is not random, which is often the case, as will be evident in the following sections, the above discussion is not entirely applicable. However, the coherence function can still be used as an indicator of the signal-to-noise ratio at each frequency bin. If the coherence function equals unity, it is always an indication that the signal-to-noise ratio is sufficient. It should be mentioned, however, that even if the coherence is unity, the estimated frequency response can be erroneous. This is the case, for example, if the *model is not true*, for instance if there is some other signal, correlated with the measured input signal, which adds to the measured output signal. An example of this case is found in Section 13.12.3.

13.8 Impact Excitation

We shall now study how the above methods are used to measure frequency responses on mechanical systems. We will start by the method of impact excitation (sometimes called impulse excitation), where an impulse hammer described in Section 7.7 is used to excite the structure under test. This technique was first developed in the 1970s, (Halvorsen and Brown 1977), and received some renewed interest in the work of Fladung (1994), who, by increasing the number of references has shown that impact testing in many cases can give high-quality results for modal analysis.

Impact excitation is easier to apply than shaker excitation, since the structure does not need to be loaded by a force transducer and mounted to a shaker. The method also has a few other advantages compared with shaker excitation, for example:

- the force is well defined in the striking direction without transverse forces (see the section on shaker excitation below),

14

Multiple-input Frequency Response Measurement

In this chapter we will introduce identification of linear systems with multiple-input signals and multiple-output signals. Such systems are found in many applications in the field of structural dynamics and acoustics. For example, the noise in different parts of the interior of a car (outputs) are made up of sound from the engine, the road, wind-induced sound, etc. (inputs). In experimental modal analysis, it is common to use several shakers to excite the test structure, and simultaneously to measure a large amount of response signals. In these cases, it is inappropriate to use the SISO estimation methods we discussed in Chapter 13, but instead frequency response functions must be estimated using a multiple-input/multiple-output (MIMO) model.

As we will show in this chapter, the only results necessary for calculating all input/output relations for linear systems, are the autospectra of all signals (input and output), and cross-spectra between each input and all other signals (including other input signals). We will use the multi-channel spectral analysis concepts introduced in Section 10.7 for the discussion of estimators for MIMO FRF models.

The success of understanding MIMO FRF estimation relies on correctly understanding the different effects of excitation signals, signal-to-noise ratios, and the extraneous noise on input, or output signals, or both these signals. We will therefore make frequent use of simulations in this chapter, to illustrate each effect without the other effects present.

14.1 Multiple-input Systems

MIMO systems are defined such that each output is caused by a linear combination of all the inputs, and there are no causal relations between any of the outputs, see Figure 14.2 where a general MIMO system is depicted. If there are causal relations between any of the outputs then one or more output signals need to be redefined as an input signal. Thus there is a linear system between each input and each output, and there can, of course, be some uncorrelated, extraneous noise added to either the inputs or outputs, or both, as for the SISO system. The MIMO system can therefore be seen as a number of parallel multiple-input/single-output systems. The fact that we have several outputs is not an important issue, from a conceptual standpoint, since each output is independent of all the other outputs. The problem, instead, lies in the decoupling of the individual contributions in a particular output, to each of the inputs. We can thus concentrate on looking at the MISO system, at least from a conceptual point of view. It may

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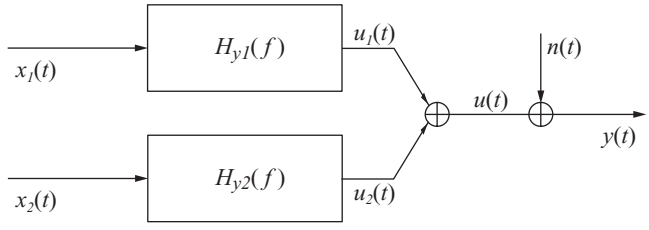


Figure 14.1 2-input/1-output system with noise on the output

be more computationally efficient to compute the entire MIMO system at once, but the understanding of the problem can be visualized by a MISO system.

14.1.1 The 2-input/1-output System

Before we introduce the general multiple-input-multiple-output (MIMO) system, we will first consider a 2-input/1-output system, as depicted in Figure 14.1 in order to visualize the concept, because the matrix algebra we use in the general case becomes rather abstract.

In Figure 14.1, the output is caused by a linear combination of the two inputs, $x_1(t)$ and $x_2(t)$ which can have some mutual correlation, but must not be completely correlated. This is no limitation, since, if the inputs are totally correlated, it is actually sufficient to use a model with only one input, as the other input can then be created by letting the first input pass a linear system. Thus the model as illustrated in Figure 14.1 is invalid in that case, since we really have a single-input system.

Some additional contaminating noise, $n(t)$, is added to the sum of the signals coming from the two inputs, $u(t)$, to produce the measured output signal, $y(t)$. In the frequency domain, the output of the system then is

$$Y(f) = X_1 H_{y1} + X_2 H_{y2} + N. \tag{14.1}$$

We obtain the least squares solution to Equation (14.1) by multiplying it by the complex conjugates X_1^* and X_2^*, respectively, and taking expected values (in a loose sense; what we do is, of course, to apply the spectral estimators from Section 10.3), which gives us the system of equations

$$\begin{cases} \mathrm{E}\left[Y X_1^*\right] = G_{y1} = \mathrm{E}\left[X_1 X_1^* H y1\right] + \mathrm{E}\left[X_2 X_1^* H_{y2}\right] + \mathrm{E}\left[N X_1^*\right] \\ \mathrm{E}\left[Y X_2^*\right] = G_{y2} = \mathrm{E}\left[X_1 X_2^* H y1\right] + \mathrm{E}\left[X_2 X_2^* H_{y2}\right] + \mathrm{E}\left[N X_2^*\right] \end{cases} \tag{14.2}$$

If we assume the noise $n(t)$ to be uncorrelated with $x_1(t)$ and $x_2(t)$, this yields

$$\begin{cases} G_{y1} = G_{11} H_{y1} + G_{21} H_{y2} \\ G_{y2} = G_{12} H_{y1} + G_{22} H_{y2} \end{cases} \tag{14.3}$$

since the cross-spectra G_{n,x_1} and G_{n,x_2} will approach zero as we take the expected values (i.e., we take many averages).

To solve the system of Equation (14.3) we could elaborate and find a way to diagonalize it to produce a system where each of the H_{y1} and H_{y2} are separated into one row. This is usually called *Gaussian elimination*, and will indeed be used in Section 14.2, as it was a historic way of solving MIMO systems. Today, however, we have good software tools for linear algebra through MATLAB/Octave and similar software. A direct matrix formulation and solution is therefore of greater interest, and we will turn to such a formulation to find the solution to Equation (14.3).

14.1.2 The 2-input/1-output System – matrix notation

Equation (14.1) can be reformulated in matrix notation as

$$Y = \lfloor H \rfloor \{X\} + \{N\} \tag{14.4}$$

where Y is a scalar because we have only one output (in general MIMO cases it will be a column vector, see Equation 14.11). Furthermore, with this formulation in the case of two inputs and one output, in Equation (14.4) $\lfloor H \rfloor$ is a row vector (1×2), equal to

$$\lfloor H \rfloor = \begin{bmatrix} H_{y1} & H_{y2} \end{bmatrix} \tag{14.5}$$

and $\{X\}$ is a column vector (2×1) equal to

$$\{X\} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \tag{14.6}$$

With these definitions Equation (14.4) is equivalent to Equation (14.1).

The least squares solution to Equation (14.4) in matrix form can now be formulated by multiplying Equation (14.4) by the matrix equivalent of the complex conjugate of X, namely the *Hermitian transpose*, see Appendix D. Instead of taking the complex conjugate alone, the Hermitian transpose, as the name implies, transposes *and* takes the complex conjugate. We then get

$$Y \lfloor X^* \rfloor = \lfloor H \rfloor \{X\} \lfloor X^* \rfloor + \{N\} \lfloor X^* \rfloor . \tag{14.7}$$

Averaging each term separately then yields the result

$$\left\lfloor G_{yx} \right\rfloor = \lfloor H \rfloor [G_{xx}] + \lfloor G_{nx} \rfloor . \tag{14.8}$$

The cross-spectrum matrix $[G_{nx}]$ in Equation (14.8) will of course approach zero in the averaging process. We therefore end up with the matrix equation

$$\left\lfloor G_{yx} \right\rfloor = \lfloor H \rfloor [G_{xx}] \tag{14.9}$$

which we then have to solve. Assuming the input cross-spectral matrix $[G_{xx}]$ can be inverted, the solution is the MISO H_1 *estimator*

$$\left\lfloor \hat{H}_1 \right\rfloor = \left\lfloor \hat{G}_{yx} \right\rfloor \left[\hat{G}_{xx} \right]^{-1} \tag{14.10}$$

where we put hats on the variables to emphasize that they are all estimates (calculated from a measurement).

There is a potential issue here: how do we know that we can invert $[G_{xx}]$? From linear algebra we know (see Appendix D) that a matrix is invertible if it has full rank, i.e., if all rows and columns are independent. This is equivalent to saying its determinant is nonzero. This is obtained in our case, if $[G_{xx}]$ is formed by averaging together several independent intermediate results $X \cdot X^*$. Two things must then be fulfilled, namely

- each element X_m in the averaging process must be *independent*, and
- more than two (in this case with two inputs) independent averages must be made when forming the matrix $[G_{xx}]$.

The first point is fulfilled if we use two independent sources driving the shakers, provided the shakers are strong enough to provide forces proportional to the voltage signals by which they are fed, see Section 14.5. The second point is hardly a problem in practice, as we will always need to take more averages than we have inputs to the system, in order to make sure the cross-spectrum matrix G_{nx} approaches zero.

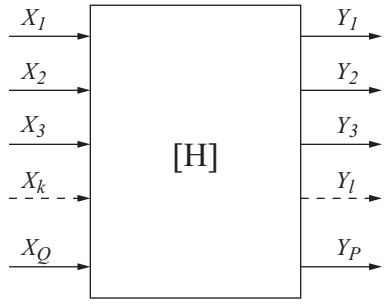


Figure 14.2 General multiple-input-multiple-output (MIMO) system. Each output signal is caused by a linear combination of the input signals, but not by any of the other output signals

14.1.3 The H_1 Estimator for MIMO

The MISO system (of which the 2-Input/1-Output was a small example) is easily extended into the general multiple-input/multiple-output, MIMO, system. For the MIMO system, any output signal is comprised of the contribution of a number of input signals. In structural dynamics (experimental modal analysis), for example, multiple-input models are used for the measurement of frequency response functions between input forces and response accelerations, when several dynamic shakers are mounted on the structure. Multiple-input models are also frequently used in noise source identification in acoustics, where dominant sources of disturbing noise are identified, see Chapter 15.

The general multiple-input/multiple-output (MIMO) system can be described as a general system with a number of input signals $x_q, q = 1, 2, \ldots, Q$, and a number of output signals y_p, $p = 1, 2, \ldots, P$, where we drop the time notation for simplicity. Each output signal is assumed to be caused only by the input signals. If there is a causal relationship between the output signals, then the model has to be redefined.

A general MIMO system is depicted in Figure 14.2. If the system is linear and time invariant, and we assume noise is contaminating only the measured output signals, we can define the system frequency response matrix $[H(f)]$ of the MIMO system so that

$$\{Y\} = [H]\{X\} + \{N\} \tag{14.11}$$

In Equation (14.11) $\{Y(f)\}$ is a column vector with dimension $(P, 1)$, $\{X(f)\}$ is a column vector with dimension $(Q, 1)$, and $[H(f)]$ is a matrix with dimension (P, Q). Thus an individual element $H_{pq}(f)$ is the frequency response between input x_q and output y_p. Row number p in $[H]$ contains the frequency responses that sum up into output signal y_p, and a column number q in $[H]$ contains the frequency responses for input signal x_q.

By multiplying Equation (14.11) by $\{X\}^H$, the Hermitian transpose of $\{X\}$, and take expected values of each term, we obtain

$$\left[G_{yx}\right] = [H]\,[G_{xx}] + [G_{nx}] \tag{14.12}$$

which is a least squares solution.

If we assume that all the noise sources in $\{N\}$ are independent of the measured input signals, the cross-spectrum matrix $[G_{nx}]$ in Equation (14.12) will be zero. Post-multiplying both sides of the equation by the inverse of $[G_{xx}]$, gives us the *MIMO H_1 estimator* as

$$\left[\hat{H}_1\right] = \left[\hat{G}_{yx}\right]\left[\hat{G}_{xx}\right]^{-1}. \tag{14.13}$$

Note that this equation has to be solved for each frequency of interest. The solution according to Equation (14.13) is possible if the matrix $[G_{xx}]$ is not singular, i.e., its determinant does not equal zero at any frequency. In theory, this means that no ordinary coherence between any two input signals can be equal to unity. In practice, as the ordinary coherence between two inputs reaches unity, there will be numerical problems with the matrix inversion.

As matrix inversion is rather inefficient computationally, in general the system in Equation (14.13) is better solved by Gaussian elimination. We will discuss solving Equation (14.13) in Section 14.1.5, but first we will define a suitable coherence function for MIMO estimation.

Example 14.1.1 *Write a MATLAB/Octave script which computes a MIMO frequency response matrix based on 2 input signals and 2 output signals. The MIMO FRF matrix will be stored in a 3D matrix similar to the spectral matrices described in Section 10.7. We assume we have already computed the input cross-spectral matrix Gxx in 3D form, and the input-output cross-spectral matrix in 3D matrix Gyx, as described in Section 10.7.2.*

The first thing we should note is that matrix inversion, using the MATLAB/Octave command ***inv****, is not recommended. Instead we will use Gaussian elimination, which in MATLAB/Octave, in this case where the matrix to invert is on the right-hand side of the multiplication matrix, is simply accomplished by the slash, '/' operator, see Appendix D. The following code then produces the H1 estimator in a variable with the same name.*

```
% First find sizes
[Nf,D,R]=size(Gyx);
% Preallocate H1
H1=zeros(Nf,D,R);
% Loop through frequencies
for f=1:Nf
  Gxxf=squeeze(Gxx(f,:,:));
  Gyxf=squeeze(Gyx(f,:,:));
  H1(f,:,:)=Gyxf/Gxxf;
end
```

End of example.

14.1.4 Multiple Coherence

If we look at the system in Figure 14.1 we realize that for multiple-input systems, the ordinary coherence functions calculated between each of the inputs and the output, will generally be less than unity. If, for example, the two input signals are uncorrelated and have approximately the same spectral densities, the two frequency response functions are of the same magnitude, and the contaminating noise is zero, each of the two ordinary coherence functions will be approximately 0.5. Thus the ordinary coherence function is not very useful for multiple-input systems.

For MIMO systems we define the *multiple coherence function* $\gamma^2_{y_p:x}$ for output signal y_p in a similar way to which the ordinary coherence function was defined for the single-input system, namely by

$$\gamma^2_{y_p:x}(f) = \frac{G_{uu}(f)}{G_{y_p,y_p}(f)}. \tag{14.14}$$

where u is the coherent output from the linear systems as in Figure 14.1.

The notation $y_p : x$ in the index is read 'y_p given (all inputs) x'. As for the single-input case, the estimation of the optimum H-systems, maximizes the power in y_p due to the input signals and puts the remaining power in the extraneous noise signal $n_p(t)$. Thus, the multiple coherence

function is interpreted, in analogy with the ordinary coherence for the single-input case, as the part of the output power spectrum G_{y_p,y_p} which is linearly dependent on any of the inputs x. As for the ordinary coherence, evidently

$$0 \leq \gamma^2_{y_p:x}(f) \leq 1 \tag{14.15}$$

where the multiple coherence function, $\gamma^2_{y_p:x} = 1$ if and only if there is no output noise, i.e., if $n_p = 0$.

For MIMO systems, there will obviously be one multiple coherence function for each output of the system. In order to find an expression which we can use in practice to estimate the multiple coherence function for output signal y_p, we formulate the coherent output spectrum, $U(f)$, from Figure 14.1, taking row p from the full matrix $[H]$, denoted $\lfloor H_p \rfloor$ by

$$U = \lfloor H_p \rfloor \{X\} + \{N\} \tag{14.16}$$

The Hermitian transpose of U, or, since it is a scalar, the complex conjugate, is

$$U^H = \lfloor X^* \rfloor \lfloor H_p \rfloor^H + \lfloor N^* \rfloor \tag{14.17}$$

Multiplying these two last equations together, and taking the expected value of each term we obtain the expression for the coherent output power spectrum G_{uu} in Equation (14.18)

$$G_{uu} = \lfloor H_p \rfloor [G_{xx}] \lfloor H_p \rfloor^H . \tag{14.18}$$

Putting the result in Equation (14.18) into Equation (14.14) we obtain the equation for the multiple coherence estimator by

$$\hat{\gamma}^2_{y_p:x} = \frac{\lfloor \hat{H}_p \rfloor \left[\hat{G}_{xx}\right] \lfloor \hat{H}_p \rfloor^H}{\hat{G}_{y_p,y_p}}. \tag{14.19}$$

We can also use the expression for the H_1 estimator in Equation (14.13) in Equation (14.19), and after some simplification we obtain

$$\hat{\gamma}^2_{y_p:x} = \frac{\left[\hat{G}_{y_p,x}\right] \left(\left[\hat{G}_{xx}\right]^{-1} \right)^H \left[\hat{G}_{y_p,x}\right]^H}{\hat{G}_{y_p,y_p}} \tag{14.20}$$

from which it is obvious that the multiple coherence, like all other results for MIMO systems, as we have noted above, can be computed from averaged auto-and cross-spectra. The estimator in Equation (14.19) is somewhat faster to compute if the frequency response matrix $\left[\hat{H}\right]$ already exists, because it does not involve any matrix inversion. Of course, if the multiple coherence is computed simultaneously with the frequency response, the inverse of the input cross-spectral matrix could be stored as an intermediate result, and then either estimator could be used with approximately the same computation effort.

Example 14.1.2 *Write a MATLAB/Octave script to produce the multiple coherence, using the data from Example 14.1.1.*

We will use the first definition in Equation (14.19) and the following code produces the multiple coherence in columns in variable Cm.

```
[Nf,D,R]=size(H);
Cm=zeros(Nf,D);
for d = 1:D    % Loop responses
    for f = 1:Nf              % Loop frequencies
        Gxxf=squeeze(Gxx(f,:,:));
        Hf=squeeze(H(f,d,:));
```

```
        Hf=Hf(:).';            % Force to row
        Cm(f,d)=real((Hf*Gxxf*Hf')/Gyy(f,d));
    end
end
```

End of example.

14.1.5 Computation Considerations for Multiple-input System

All the discussions above assume that the input signals are not completely coherent. This in principle means that the signals used to average together the input- and output auto- and cross-spectral matrices have to be formed by independent averages. This leads to the conclusion that the data must, in general, not be periodic, since a sine wave is completely coherent with any other sine wave of the same frequency. We will present two exceptions to this later in this chapter; (i) so-called *periodic random*, see Section 14.4.3, and (ii) the multiphase stepped-sine procedure, see Section 14.4.4. For the general case, however, the input signals must not be periodic. Transient signals will in general not be valid either, except for the case where the transients consist of (independent) burst random signals.

A direct effect of the procedure of conditioning the input signals as described above, is that the number of averages used for calculation of the spectral matrices, has to be larger than the number of inputs. This fact is analogous with the single-input case, where the coherence function is not defined for the first average, as it would equal unity, even with no actual correlation between the input and output signals. Similarly, with several inputs, enough statistically independent input records have to be combined into the auto- and cross-spectra, in order for all the ordinary coherence functions between the inputs to be well defined.

This requirement is of course also necessary when using matrix inversion to solve a MIMO system as in Equation (14.13). In linear algebra terms, the rank of the input cross-spectral matrix $[G_{xx}]$ equals unity after one average, two after two averages, and so on until the number of averages is larger than or equal to the number of inputs, after which the rank of the matrix equals the number of inputs (which is also the size of the matrix $[G_{xx}]$).

14.1.6 The H_v Estimator

When estimating MIMO systems on mechanical systems, the H_1 estimator is often not optimal. In this application the MIMO system is always defined as response over force, and the H_1 estimator will optimize for contaminating noise in the acceleration (output) signals. Around the natural frequencies of the structure, however, the acceleration is large and thus the extraneous noise is small relative to the acceleration signal. At the same frequencies, the force signals are often low, due to limitations in the excitation system (shaker, amplifier, and stinger), which makes it difficult to excite the structure. Thus, the inputs of the MIMO system, the forces, are more prone to be contaminated by extraneous noise than the outputs, which calls for an estimator equivalent to the H_2 estimator for the single-input case. This proves tricky, however, as it can easily be shown that an H_2 estimator for multiple-input systems can be defined only for the special case where the number of response signals equals the number of forces, which is, of course, not generally the case. Because the H_2 estimator for MIMO systems is of limited interest, we leave the proof of this fact to Problem 14.2.

The H_v *estimator* was developed in the early 1980s (Mitchell 1982; Rocklin *et al*. 1985) to yield better results in general cases with contaminating noise on both input and output. The H_v estimator assumes that all the extraneous noise sources are incoherent with the true input and output signals. Then, the linear system is defined by the matrix equation

$$\{N\} = [H]\{X\} + \{Y\} \tag{14.21}$$

where $\{n\}$ is an error vector consisting of all extraneous noise. Next we form the *error autospectrum matrix*, G_{nn} by multiplying Equation (14.21) by its complex conjugate and taking the expected value. This results in

$$[G_{nn}] = \left[G_{yy}\right] + [H][G_{xx}][H]^H - [H]\left[G_{xy}\right] - \left[G_{yx}\right][H]^H . \tag{14.22}$$

Equation (14.22) can be shown to be equal to the composed matrix

$$[G_{nn}] = \left[\, I \mid H \,\right] \begin{bmatrix} G_{yy} & G_{yx} \\ G_{xy} & G_{xx} \end{bmatrix} \begin{bmatrix} I \\ H^H \end{bmatrix} \tag{14.23}$$

Equation (14.23) is decomposed by the eigenvalue decomposition

$$[G_{nn}] = [U]^H [\Lambda][U] \tag{14.24}$$

where the matrix $[U]$ is the eigenvector matrix, and $[\Lambda]$ is a diagonal matrix containing the eigenvalues on the diagonal. It can be shown with some advanced linear algebra, that the solution to minimizing the trace of Equation (14.24), which is a so-called *Rayleigh quotient*, is found by choosing the eigenvector corresponding to the lowest eigenvalue in $[\Lambda]$. This is the H_v estimator. For the single-input/single-output case, the H_v estimator reduces to the geometric mean of the H_1 and H_2 estimators

$$\hat{H}_v = \frac{G_{yx}}{|G_{yx}|}\sqrt{\frac{G_{yy}}{G_{xx}}} = \sqrt{\hat{H}_1 \hat{H}_2}. \tag{14.25}$$

14.1.7 Other MIMO FRF Estimators

It can be shown (White *et al*. 2006) that the H_v estimator in Section 14.1.6 is a special case of a general maximum likelihood (ML) estimator, for the case where the input and output noise at each frequency are equal. It can also be shown that the minimization problem for noise on both input and output is only soluble for general random inputs, if at least the ratio of the two noises are known, which is rarely the case in our applications. The H_v estimator, is thus not a very good estimator for our purposes, since we can assume on mechanical systems with high dynamic range, that at the critical frequencies (typically the natural frequencies, and perhaps to some extent the antiresonances), we have dominant extraneous noise in *either* of the two signals (input or output), but not in both simultaneously. It therefore seems as though we are in vain trying to find any better estimator than the H_1 estimator, for the general case with random inputs.

In addition to the H_1 and H_v estimators, which are the estimators commonly available in commercial software for noise and vibration analysis, several other estimators exist, which can sometimes be alternatives if dedicated software is developed. The reason to use such an alternative estimator has to be based on the fact that we want to minimize the bias error in the FRF due to noise on the input signals in addition to the noise on the output signals, because otherwise the H_1 is a superior estimator. We will therefore present references to some alternatives that can be used in such cases.

One alternative is the H_c estimator described in Section 13.2.3, which also be defined for MIMO situations. However, the price is that one extra measurement channel per force is allocated, which makes it less attractive. A second approach is the H_α estimator suggested in Antoni *et al.* (2004) which makes use of theory of cyclostationarity to find an estimator which is asymptotically unbiased even in the case of input noise. A third alternative is to use time domain averaging and periodic excitation signals. This alternative will be demonstrated in Section 14.5.3.